Jason da Silva Castanheira
London, United Kingdom
j.castanheira@ucl.ac.uk

Academic Positions

Postdoctoral Research Fellow at the division for Psychology and Languages Sciences, University College London — April 2024-present
Supervisor: Prof. Stephen Fleming

Education

PhD in Neuroscience
MEG, myself, and I: Understanding inter-individual diversity of neurophysiological brain activity — Sept 2018 – April 2024
McGill University, Montreal, Quebec
GPA: 4.0; supervisor: Prof. Sylvain Baillet

Bachelor of Science: major in Neuroscience; minor in Psychology — Sept 2014 – May 2017
McGill University, Montreal, Quebec
GPA: 3.97

Teaching Experience

Guest Lecturer for CLNE0065: Human Perception and Cognition, UCL — Oct 2025 & 2026

Introduction to EEG/MEG Analysis: An Online Methods Workshop Series, UCL — March-May 2026

Introduction to R & statistics workshops Graduate Student Association for Neuroscience (GSAN), McGill University — March-May 2021

Introduction to MATLAB & Signal Processing Graduate Student Association for Neuroscience (GSAN), McGill University — March-May 2020

Selected Publications

Published first author manuscripts

da Silva Castanheira, J., He, C., Shea, N., & Fleming, S. (2026) *How attention simplifies mental representations for planning*. eLife.

da Silva Castanheira, J., Wiesman, A., Taylor, M., & Baillet, S. (2025) *The lifespan evolution of individualized neurophysiological traits*. Cell Reports.

da Silva Castanheira, J., Poli, J., Hansen, J., Misic, B., & Baillet, S. (2025) *Genetic foundations of interindividual neurophysiological variability*. Science Advances.

*Landry, M., *da Silva Castanheira, J, Raz, A., Baillet, S., Sackur, J. (2024) *A lateralized alpha-band marker of the interference of exogenous attention over endogenous*. Cerebral Cortex.

da Silva Castenheira, J., Wiesman, A., Hansen, J., Misic, B., Baillet, & PREVENT-AD, QPN, (2024) *The neurophysiological brain-fingerprint of Parkinson's disease*. eBioMedicine.

da Silva Castanheira, J.*, Perez, H. O.*, Misic, B., & Baillet, S. (2021) *Brief segments of neurophysiological activity enable individual differentiation*. Nature Communications.

Published co-authored manuscripts

Baracchini, G., et al. (2026) *The biological role of local and global fMRI BOLD signal variability in multiscale human brain organization.* Nature Communications

Mokri, E., Landry, M., da Silva Castanheira, J., & Mendola, J. (2026) *Neural Markers of Interocular Grouping During Binocular Rivalry With MEG.* Human Brain Mapping.

Mokri, E., da Silva Castanheira, J., Baillet, S., & Mendola, J. (2025) *Alpha oscillations mediate inhibitory control and perceptual transitions during binocular rivalry*. iScience.

Wiesman, A., da Silva Castanheira, J., Fon, E., Baillet, S., PREVENT-AD Research Group, & Quebec Parkinson Network. (2024) *Alterations of Cortical Structure and Neurophysiology in Parkinson's Disease Are Aligned with Neurochemical Systems.* Annals of Neurology.

Wiesman, A., da Silva Castanheira, J., Degroot, C., Fon, E., Baillet, S., PREVENT-AD Research Group, & Quebec Parkinson Network. (2023) *Adverse and compensatory neurophysiological slowing in Parkinson's disease*. Progress in Neurobiology.

Mokri, E., da Silva Castanheira, J., Sidrah, L., Landry, M., & Mendola, J. (2023) *Effects of interocular grouping demands on binocular rivalry*. Journal of Vision.

Landry, M., da Silva Castanheira, J., & Jerbi, K. (2023) *Differential and overlapping effects between exogenous and endogenous attention shape perceptual facilitation during visual processing*. Journal of Cognitive Neuroscience.

Bock, E., Fesi J., da Silva Castenheira, J., Baillet, S., Mendola, J. (2023) *Distinct dorsal and ventral streams for binocular rivalry dominance and suppression revealed by magnetoencephalography*. European Journal of Neuroscience.

Wilson, L., da Silva Castanheira, J., Baillet, S. (2022) *Time-resolved parameterization of aperiodic and periodic brain activity.* eLife.

Wiesman, A. I., **da Silva Castanheira, J.**, & Baillet, S. (2022) *Stability of spectral estimates in resting-state magnetoencephalography: Recommendations for minimal data duration with neuroanatomical specificity*. NeuroImage.

Landry, M., **da Silva Castanheira, J.**, Milton, D., & Raz, A. (2022) *Suggestion alters Stroop automaticity: hypnotic alexia through a proactive lens*. Psychology of Consciousness.

Duclos, C. et al. (2021) *Differential classification of states of consciousness using envelope- and phase-based functional connectivity*. NeuroImage.

Landry M., **da Silva Castanheira, J.**, Sackur, J., & Raz, A. (2021) *Investigating how the modularity of visuospatial attention shapes conscious perception using type 1 and type 2 signal detection theory*. Journal of Experimental Psychology: Human Perception and Performance.

Landry, M., **da Silva Castanheira, J.**, Sackur, J., & Raz, A. (2021) *Difficult turned easy: suggestion renders a challenging task simple.* Psychological Science.

Motta-Ochoa, R., Bresba, P., **da Silva Castanheira, J.**, Lai Kwan, C., Shaffer, S., Julien, O., William, M., & Blain-Moraes, S. (2021). *"When I hear my language, I travel back in time and I feel at home": Intersections of culture with social inclusion and exclusion of persons with dementia and their caregivers*. Transcultural Psychiatry.

Conference papers

da Silva Castenheira, J., He, C., Shea, N., & Fleming, S. (2026) *Imagining internal mental representations for planning*. Proceedings of the Annual Meeting of the Cognitive Science Society.

Manuscripts in Review

*Wilson, L., ***da Silva Castenheira, J.,** Lévesque Kinder, B., & Baillet, B. (in review Imaging Neuroscience) *A Bayesian Model-Selection Approach for Determining the Number of Spectral Peaks in Neural Power Spectra*.

da Silva Castenheira, J., Landry, M., & Fleming, S. (in review Imaging Neuroscience) *Analytical Choices Impact the Estimation of Rhythmic and Arrhythmic Components of Brain Activity*.

Lapatrie, M., **da Silva Castenheira, J.,** Aydin, I., & Baillet, B. (in review Nature Communications) *Unsupervised Representation Learning Generates Differentiable Neurophysiological Profiles.*

Prepared manuscripts

da Silva Castanheira, J., Chang, C., O'Neill, J., Liang, X., Xiao, J., Shea, N., & Fleming, S. (In prep) *Rapid perceptual reconfiguration of people's world models*.

* authors contributed equally

Fellowships and Scholarships

Natural Sciences and Engineering Research Council of Canada (NSERC) Canada Postdoctoral Fellowship (CAD $70,000 for 2 years)	April 2024
Fonds de Recherche du Québec Nature et technologies (FRQNT) Postdoctoral Fellowship (CAD $45,000 for 2 years) —Declined	April 2024
Natural Sciences and Engineering Research Council of Canada (NSERC) Canada Graduate Scholarships-Doctoral (CAD $35,000 for 3 years)	April 2021
Fonds de Recherche du Québec Nature et technologies (FRQNT) Doctorate Scholarship (CAD $84,000)	April 2020
Fonds de Recherche du Québec Nature et technologies (FRQNT) Masters Scholarship (CAD $17,500)	April 2019
NSERC CREATE Graduate Scholarship (CAD $18,000)—Declined	April 2018
NSERC Canada Graduate Scholarships-Masters (CAD $17,500)	April 2018
Recruitment Award IPN	Sept 2018
FRQNT Supplement of the NSERC Undergraduate Summer Research Award (CAD $2,000)	Summer 2017
NSERC Undergraduate Summer Research Award (CAD $5,625)	Summer 2017

Invited Talks

MEGUKI Meeting (Nottingham, UK). Tracking the neural dynamics of simplified mental representations	Sept 2026
UCL LGBTQ+ Science Research Showcase, UCL (London, UK). Internal attention selects task-relevant information for planning	June 2026
Invited talk, Université Laval (Québec City, Canada). Visuospatial attention guides simplified mental representations for planning	March 2026
UCL-EEG Community methods meeting (London, UK). Rapid Invisible Frequency Tagging to Track the Neural Dynamics of Subjective Experiences	Jan 2026
UCL LGBTQ+ Science Research Showcase, UCL (London, UK). How attention simplifies mental representations for planning	June 2025
Brain Meeting, UCL (London, UK). Inter-individual diversity in brain function and conscious experience	Oct 2024
Consciousness & Computation Conference, UCL (Zagreb, Croatia). Brain-fingerprinting: unravelling the biological origins of inter-individual differences	April 2024

Cutting Gardens 2023: Conference on cutting edge methods in M/EEG (Los Angles, USA). The neurophysiological brain-fingerprint of Parkinson's disease	Sept 2023
Wisdom Exchange Project: science communication initiative where researchers explain cutting edge research to older adults. Brain-fingerprints Change Across the Lifespan	June 2023
Chinese Open Science Network: online symposia to promote open science. An introduction to SPRiNT: Spectral Parameterization Methods.	Oct 2022
Conference: Society for Applied Neuroscience, Thessaloniki, Greece. Neurophysiological fingerprints of motor and cognitive decline in Parkinson's Disease.	Sept 2022
22nd International Conference on Biomagnetism, Birmingham, UK. MEG, myself, & I: brief segments of neurophysiological activity enable individual differentiation.	Aug 2022
Brenda Milner Neuropsychology Research Day, The Neuro. More than just a fingerprint: Neurophysiological differentiability relates to motor and cognitive decline in Parkinson's disease.	May 2022
McGill-NIPS Joint Symposium, Virtual symposia to promote international collaborations between universities. MEG, myself, & I: brief segments of neurophysiological activity enable individual differentiation.	Jan 2022
NeuroLingo, a science communication initiative where scientists present the research in a *'TED talk'*-like presentation. Individual identification from brain activity: NeuroLingo	July 2021
Cognitive Research at McGill Brownbag Talks: Department of Psychology, McGill University. The neural dynamics of anticipation: how temporal predictions boost conscious perception	Jan 2019

Conference Posters

da Silva Castanheira J. *, Perez, H. O., Misic, B., & Baillet, S. (2019) *Identifying individuals from resting-state MEG*. Poster at the IPN retreat, Montreal, Canada.

da Silva Castanheira J.*, Landry M, Fleming S (2026). Poster: Quantifying Rhythmic and Arrhythmic Components of Brain Activity. Annual meeting of the Organization for Human Brain Mapping (OHBM), Bordeaux, France.

da Silva Castanheira J.*, He C, Shea N, Fleming S (2026). Poster: The Neural Dynamics of Simplified Mental Representations. Annual meeting of the Organization for Human Brain Mapping (OHBM), Bordeaux, France.

da Silva Castanheira J.*, He C, Shea N, Fleming S (2026). Poster: Imagining internal mental representations for planning. Cognitive Science Society (CogSci), Rio de Janeiro, Brazil.

da Silva Castanheira J.*, Shea N, Fleming S (2025). Poster: Recommendations for Quantifying Rhythmic and Arrhythmic Components of Brain Activity. MEGUKI, London, UK.

da Silva Castanheira J.*, Shea N, Fleming S (2025). Poster: How attention simplifies mental representations for planning. Computational Cognitive Neuroscience (CCN), Amsterdam, Netherlands.

da Silva Castanheira J.*, Wilson L, Baillet S (2023). Poster: Bayesian parameterization of the neurophysiological spectrum. Annual meeting of the Organization for Human Brain Mapping (OHBM), Montreal, Canada.

da Silva Castanheira J.*, Wilson L, Baillet S (2023). Poster: Bayesian parameterization of the neurophysiological spectrum. Annual meeting of the Organization for Human Brain Mapping (OHBM), Montreal, Canada.

da Silva Castanheira J. *, Wiesman AI, & Baillet S (2023). Poster: Neurophysiological fingerprinting differs in content but not accuracy across the adult lifespan. Cognitive Neuroscience Society, San Francisco, USA.

da Silva Castanheira J. *, Wiesman AI, Hansen JY, Misic B, Baillet S, PREVENT-AD Research Group, Quebec Parkinson Network (2022). Poster: More than just a fingerprint: Neurophysiological differentiability relates to motor and cognitive decline in Parkinson's Disease. 22nd International Conference on Biomagnetism, Birmingham, UK.

da Silva Castanheira J.*, Wiesman AI, Baillet S, PREVENT-AD Research Group, Quebec Parkinson Network (2022). Poster: Neurophysiological fingerprints of Parkinson's Disease. Annual meeting of the Organization for Human Brain Mapping (OHBM), Glasgow, UK.

da Silva Castanheira J. *, Perez, H. O., Misic, B., & Baillet, S. (June 2020) *Identifying individuals from resting-state MEG*. Poster at Organization for Human Brain Mapping (OHBM), virtual conference.

da Silva Castanheira, J.*, & Blain-Moraes, S. (2018) *Contrasting amplitude and phase-based measures of functional connectivity across states of consciousness*. Poster at NeuroSymposium, Montreal, Canada.

Mentorship

Master's thesis project, UCL: Jiong Xiao & Xinyue Liang	Oct 2025- Aug 2026
Research Assistant, UCL: Christina Chang He	Sept 2025- Aug 2026
Master's thesis project, UCL: Christina Chang He	Oct 2024- Aug 2025
Research Internship CentraleSupélec, Université Pairs-Saclay: Jonathan Poli	Summer 2023
Undergraduate research project, McGill University: Monica Song, Le Thuy Duong Nguyen & Benjamin Lévesque Kinder	Fall 2022- Winter 2023
Undergraduate research project, McGill University: Antonia Panaitescu & Phan Dang	Sept 2020 - April 2021
Undergraduate research project, McGill University: Luc Wilson	Jan - May 2020
Undergraduate research project, McGill University: James Garneau, Anais Durand & Jessie Bull	Sept 2019 - April 2020
Medical research project—National University of Colombia: Alejandro Gamba S.	March - July 2019
Undergraduate research project, McGill University: Marlee Brownstein, Amanada Gossack, Chelsea Osei, & Christophe Tanguay Sabourin	Sept 2018 - April 2019

Awards & Achievements

Governor General's Gold Medal for outstanding doctoral thesis in the Natural Sciences	May 2024
Gordon A. Maclachlan Prize for outstanding doctoral thesis in the biological and health sciences	May 2024
Neuro Travel Award	March 2023
CRBLM Travel Award	Sept 2022
PGSS Travel Award	Sept 2022
BioMag Student Travel Award	Aug 2022
GREAT Travel Award	June 2022
CIHR Brain Star Award	May 2022

Engagement with Scientific Community

Ad-hoc reviewer at the following journals: *PLoS computational biology, NeuroImage, Neurobiology of Aging, Nature Mental Health, IEEE Journal of Biomedical Informatic, Science Advances, European Journal of Neuroscience, Imaging Neuroscience, Open Mind: Discoveries in Cognitive Science, Scientific Reports, Journal of Vision, PNAS Nexus, Network Neuroscience & Cerebral Cortex*

CIFAR Neuroscience of Consciousness Winter School — Dec 2025

Volunteer at the XXI World congress of Medical and Clinical Hypnosis — Aug 2018

Public Engagements

TV feature of Fleming's Lab: ARTE ZDF, *Intuition & Gut Feeling in the Age of AI* — Oct 2026

TV feature of Fleming's Lab: *NANO Doku, Forschung extrem – Unser Weltbild am Limit?* — June 2025

Wisdom Exchange Program: intergenerational social connection and knowledge exchange program — Jan 2021- April 2024

Brain Info: science communications initiative (administration role) — Jan 2021- April 2024

NeuroLingo: neuroscience talks for the general public (graphic designer) — Sept 2022- Sept 2023

IPN Buddy Program: mentoring incoming Neuroscience Graduate students — Sept 2019- Aug 2023

Cognitive Neuroscience research day for the SHAD empowering youths program — July 2018 & 2019

Interfacing Biomusic with Autism: workshops on neuroscience of emotion at *Spectrum Productions* — April 2018

Volunteer at the Eureka! Science Festival for Children — June 2015

Languages & Skills

Spoken languages: English, Portuguese, and French

Written languages: English, Portuguese, and French

Programming languages: Bash, R, MATLAB, and python

Software: Brainstorm, BrainVision Analyzer, EEGLab, RStudio, Jupyter Notebooks, Eyelink 2000, Inkscape, Microsoft Office (Word, PowerPoint, and Excel), Unix Operating Systems

Hardware: Electrophysiological acquisition with NetStation EGI and BrainVision Actichamp. Certified Magnetoencephalography CTF user.